Linkan Dash

Computational Biologist | Data Analyst

linkandash15@gmail.com   linkangit.github.io/portfolio   github.com/linkangit

515-817-3077   Piscataway, New Jersey

Professional Summary

Highly motivated scientist with demonstrated expertise in data engineering, data management strategies, and computational data analysis. Offering 2+ years of industry R&D, 8+ years of molecular wet-lab, and 6 years of bioinformatics experience. Specialized in functional genomics (with focus on NGS, sequencing platforms, and genotyping), chromatin architecture, plant epigenomics, and innovative problem-solving through statistical computing (R, Python). Proven ability in cross-functional collaboration, project leadership, and delivering impactful data visualizations and reporting solutions to support research and development pipelines. Self-driven hybrid scientist with strong organizational skills, critical thinking, and adaptability to change. Actively seeking roles in Data Analytics, Computational Biology, Bioinformatics, or Genomics R&D.

Education

Iowa State University | Ames, Iowa — Aug 2018 – May 2023

Ph.D. in Genetics and Genomics **NISER** | Bhubaneswar, India — Aug 2013 – May 2018

B.S./M.S. Dual Degree in Biological Sciences

Work Experience

Postdoctoral Associate – Chromatin Architecture & Gene Regulation — Jan 2026 – Present

Rutgers University, Dr. Mark Zander's Lab | Piscataway, New Jersey

- **Advanced chromatin architecture analysis:** Employing cutting-edge Micro-C-XL ChIP and Hi-C-ChIP seq technologies to investigate 3D chromatin looping mechanisms during Jasmonic acid signaling in crop and medicinal plants
- **Computational-intensive data analysis:** Designing and implementing custom bioinformatics pipelines for processing and analyzing high-resolution chromatin interaction datasets, integrating multi-omics data to elucidate gene regulatory mechanisms
- **Hybrid wet-lab and computational expertise:** Leveraging combined computational and molecular biology skills to execute end-to-end experimental workflows from library preparation to advanced data analysis and interpretation
- **Innovative problem-solving:** Applying statistical computing tools (R, Python) and deductive reasoning to identify chromatin looping patterns and their role in stress-responsive gene regulation
- **Cross-functional collaboration:** Working collaboratively with plant biologists and computational scientists to advance understanding of chromatin dynamics in plant defense responses

Postdoctoral Scientist – Developmental Genomics & Mitochondrial Physiology — Sep 2025 – Dec 2025

Marine Biological Laboratory | Woods Hole, Massachusetts

- **Data quality monitoring & analysis:** Designed and executed time-course bulk RNA-seq across critical developmental windows (1, 2, 4, 6, 8, 16, 32 cell stage embryos) in *Brachionus manjavacas*. Managed and manipulated large datasets with rigorous quality control to identify stage-specific transcriptional networks
- **Innovative reporting solutions:** Built end-to-end analysis workflows applying statistical computing tools to identify developmental metrics and candidate genes. Utilized deductive reasoning and critical thinking to prioritize 5 candidate genes showing 2-fold reduced expression in high maternal-age embryos
- **Advanced data visualization:** Generated high-quality microscopy data and performed nuclear staining using Hoechst dyes. Applied data visualization techniques through confocal and fluorescence microscopy to support transcriptomic findings and elucidate cellular mechanisms
- **Independent and collaborative work:** Demonstrated ability to work independently while executing functional validation via optimized RNAi protocols, showcasing experimental design from hypothesis to functional characterization

- **Cross-functional contributions:** Integrated transcriptomic analysis with functional assays to establish gene-to-phenotype relationships, applying innovative approaches to understand maternal age effects at molecular and cellular levels

Postdoctoral Scientist – Genome Center of Excellence Aug 2023 – Aug 2025
Corteva Agriscience | Johnston, Iowa

- **Project leadership & data management:** Independently led two high-priority R&D bioinformatics projects from design to completion, including pipeline development and optimization for biomarker discovery. Formed partnerships with cross-functional sequencing and genotyping lab teams to identify solutions and improvements to support metrics and data quality, achieving 100% on-time delivery
- **Innovative data engineering:** Engineered a novel computational tool development workflow for mapping random T-DNA integration sites using variant calling pipelines. Reduced assay costs by 60% through creative approaches to solve complex problems, aligning with organizational goals of scalability and cost-efficiency
- **Multi-omics innovation & data visualization:** Designed and implemented deep learning-assisted single-cell (scRNA-seq/scATAC-seq) and Visium spatial transcriptomics workflows. Built impactful data visualizations and integrated multi-omics datasets to uncover gene regulatory mechanisms and epigenome dynamics, enabling predictive insights for complex trait biology
- **Advanced reporting & database management:** Built end-to-end analysis workflows that applied machine learning algorithms to bulk RNA-seq datasets. Demonstrated ability to work with relational databases and query internal data for reporting purposes, resulting in discovery of >30 novel candidate genes from SE-centric gene regulatory network modules
- **Technology adaptation & solution delivery:** Adapted advanced live-cell imaging approaches to monitor contamination-sensitive embryogenic tissues. Demonstrated flexibility and rapid uptake of new technologies, establishing protocols now adopted across multiple teams
- **High-throughput platform optimization:** Optimized automated image-based phenotyping of somatic embryos across plant species, improving throughput, reproducibility, and standardization of critical developmental assays under various stress conditions, supporting monitoring of genomics sequencing and genotyping platforms
- **Technical communication & cross-functional collaboration:** Communicated complex technical information to a variety of audiences in a clear and concise manner. Provided technical expertise to teams spanning molecular biology, breeding, and data science. Published peer-reviewed review on crop plant epigenomics and contributed key datasets to patent filings
- **Leadership & change management:** Served as elected ASPB Postdoc Ambassador and Council member (2025–26), strengthening visibility of Corteva's science. Demonstrated ability to lead and drive change and improvements through teams across the organization. Actively contributed to DEI initiatives, fostering inclusive science communication

Graduate Research Assistant Aug 2018 – May 2023
Iowa State University | Ames, Iowa

- **Independent project leadership & data management:** Conceived and drove four major research projects in plant stress physiology and auxin signaling, securing >$1M in NSF grant funding. Published four first- and co-authored peer-reviewed manuscripts, demonstrating strong people and project leadership skills
- **Data engineering & pipeline development:** Built custom genomic data analysis pipelines in R for RNA-seq, ChIP-seq, and DAP-seq datasets. Enhanced quality control and reproducibility by 40%, enabling high-confidence discovery of transcription factor binding and gene regulatory mechanisms
- **Statistical computing & large dataset manipulation:** Applied knowledge of R, Python and related programming and scientific computing tools to manage and manipulate large datasets. Developed innovative solutions for transcriptomic and epigenomic data integration
- **Data visualization excellence:** Produced >50 publication-quality figures using ggplot2 and Tidyverse. These impactful data visualizations were widely used in manuscripts, presentations, and grant proposals, communicating complex technical information effectively

- **Cross-disciplinary collaboration & problem-solving:** Worked effectively in a team environment, collaborating with molecular biologists, computational scientists, and physiologists. Applied deductive reasoning and critical thinking to design cutting-edge experiments, resulting in two novel methodologies
- **Understanding of genomics platforms:** Optimized >50 molecular biology protocols (CRISPR constructs, NGS library preparation, protoplast transfections, qPCR/ddPCR), reducing error rates and increasing throughput by 25%. Demonstrated understanding of genomics, sequencing, genotyping, and gene editing platforms
- **Time management & organizational skills:** Capable to work independently and collaboratively while applying good time management skills. Managed zebrafish stem-cell cultures alongside plant experiments, demonstrating adaptability across model systems
- **Mentorship & communication:** Organized six high-profile departmental seminars with leading plant biologists. Provided informal mentorship to junior researchers, demonstrating strong interpersonal communication skills

Undergraduate Research Assistant Aug 2016 – May 2018
NISER | Bhubaneswar, India

- **Data quality & laboratory excellence:** Led two molecular biology studies, conducting >100 RT-PCR/qPCR, Western blot, and cloning assays with 98% success rate, demonstrating attention to data quality
- **Team leadership & collaboration:** Mentored a five-member research team, focusing on applied research, knowledge sharing, and detailed record-keeping. Improved team efficiency by 15% through strong teamwork skills and well-organized approach to project management
- **Technical reporting & communication:** Authored protocols, technical reports, and proof-of-concept findings; presented at two national conferences, demonstrating ability to communicate findings in a clear and concise manner
- **Data collection & monitoring:** Managed greenhouse operations with emphasis on experimental design, data collection, and equipment performance monitoring for plant biology research, increasing crop yield by 10%

Relevant Skills

Data Engineering & Management: Custom pipeline development, computational workflow engineering, data management strategies, aligned data solutions, relational database querying (SQL), large-scale data manipulation, quality control frameworks
Statistical Computing & Programming: R (tidyverse, ggplot2, DESeq2, WGCNA), Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), Bash/Shell scripting, Machine learning algorithms (Random Forest, KNN, Regression, Naive Bayes), HPC (LSF/bsub), Git, Snakemake
Data Visualization & Reporting: ggplot2, Matplotlib, Seaborn, JMP, GraphPad Prism, Loupe Browser, impactful data visualization development, publication-quality figure generation, reporting tools for R&D pipelines
Genomics & Sequencing Platforms: NGS (Illumina RNA library prep), sequencing platforms (10x Genomics Chromium, Visium), genotyping platforms, single-cell/nuclei isolation, variant calling (SNVs, indels), GATK, SAMtools, BEDtools, STAR, Micro-C-XL ChIP, Hi-C-ChIP seq
Bioinformatics & Data Analysis: Multi-omics integration (SC-ION), gene regulatory network analysis (GENIE3, WGCNA), transcription factor binding analysis (ChIP-seq, DAP-seq, ATAC-seq), scRNA-seq/scATAC-seq, spatial transcriptomics (Cellranger, Spaceranger, Scanpy, Seurat v5), trajectory analysis, chromatin interaction analysis
Molecular Biology & Gene Editing: CRISPR/Cas9 editing (SDN1/2, CRISPRa/i), gene editing platforms, cloning, PCR/qPCR/ddPCR, transformation protocols, protein purification/QC
Tool Development: Innovative computational tool development, workflow engineering, solution identification and implementation, pipeline optimization
Database & Cloud: Relational databases, extract, summarize, and report data, Azure cloud computing, Microsoft Office Suite
Microscopy & Imaging: Confocal/fluorescence microscopy, live cell imaging, image analysis (ImageJ/FIJI), subcellular localization, BiFC assays

AI & Advanced Tools: ChatGPT, DeepSeek, AlphaFold3 (in silico protein modeling)
Leadership & Soft Skills: Strong organizational skills, project leadership, cross-functional collaboration, drive change and improvements through teams, critical thinking, deductive reasoning, innovative mindset, adaptability to change, ability to work independently and collaboratively, time management, strong interpersonal communication skills, willingness to help others, desire to excel

Certificates

- Python for Data Science, AI and Development, IBM (April 2025)
- Genomic Data Science and Clustering, UC San Diego (May 2025)

Awards

- Top Cited Article in Plant Direct Journal (2021–2022)
- Research Excellence Award (May 2023), Iowa State University, Best Doctoral Thesis Award, Spring 2023
- Deutscher Akademischer Austauschdienst German Academic Exchange Fellowship (May 2015 – Aug 2015)
- DST-INSPIRE Fellowship, Department of Science and Technology, Government of India
- National Science Olympiad (2009 – 2011)

Leadership Roles

- Member of the ASPB Council – Early Career Representative (July 24th 2025 – Present)
- Ambassador (Jan 2025 – Present), American Society of Plant Biologists (ASPB)
- Member of Business Resource Group (Sep 2023 – Aug 2025), Corteva Agriscience, Growing Asian Impact Network (GAIN)
- Head of Social Team (Jan 2022 – May 2023), Iowa State University, Grad Student and Post-doc Organization, GDCB Department

Publications

- Hu, Y., **Dash, Linkan**, May, G., Sardesai, Nagesh, & Deschamps, Stephane (2024). Harnessing Single-Cell and Spatial Transcriptomics for Crop Improvement. *Plants*, 13(24), 3476.
- **Dash, Linkan**, Swaminathan, Sivakumar, Šimura, Jan, et al. "Changes in cell wall composition due to a pectin biosynthesis enzyme GAUT10 impact root growth." *Plant Physiology*, 2023.
- **Dash, Linkan**, McReynolds, Maxwell R, et al. "The Auxin Response Factor ARF27 is required for maize root morphogenesis." *bioRxiv*, 2023.
- Cowling, Craig L*, **Dash, Linkan***, Kelley, Dior R. "Roles of auxin pathways in maize biology." *Journal of Experimental Botany*, 2023.
- McReynolds, Maxwell R*, **Dash, Linkan***, et al. "Temporal and spatial auxin responsive networks in maize primary roots." *Quantitative Plant Biology*, 2022.
- **Dash, Linkan**, McEwan, Robert E, et al. "slim shady is a novel allele of PHYTOCHROME B present in the T-DNA line SALK_015201." *Plant Direct*, 2021.
- Patel, Bhavika B, Clark, Kendra L, **Dash, Linkan**, et al. "Isolation and culture of primary embryonic zebrafish neural tissue." *Journal of Neuroscience Methods*, 2019.

**Co-first authors (authors have contributed equally to the manuscript)*